COHOMOLOGY SHEAVES OF A DERIVED PUSHFORWARD AS A SHEAFIFIED PRESHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the displayed description and the accompanying identity.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived pushforward along a morphism of ringed sites and the computation of its cohomology sheaves. Same chapter, the description of Rf_* through K-injective representatives and of the section functors over objects. Supplies the object whose cohomology sheaves are described.
- Identification of sections over V with sections over u(V). Same chapter together with the companion lemma on the Leray identification, which gives H^i(V, Rf_*K) = H^i(u(V), K). Supplies the second displayed identity and the sheafification step.

Source reference. Original lemma 0D6H: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4403–4403. Navigation pointer invoked at proof line 4448; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BKV: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4349–4349. Navigation pointer invoked at proof line 4449; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let $f : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed sites corresponding to the continuous functor $u : \mathcal{D} \to \mathcal{C}$. Let K be in $D(\mathcal{O}_\mathcal{C})$. Then $H^i(Rf_*K)$ is the sheaf associated to the presheaf*

$$V \mapsto H^i(u(V), K) = H^i(V, Rf_*K)$$

Proof. The equality $H^i(u(V), K) = H^i(V, Rf_*K)$ follows upon taking cohomology from the second statement in Lemma 0D6H. Then the statement on sheafification follows from Lemma 0BKV. □